

day7 Notebook

Cell 1 (markdown)

Diabetes Risk Predictor: A Mini ML Project

Cell 2 (markdown)

This notebook aims to develop a machine learning model to predict diabetes risk based on health indicators. We will be using the 'Diabetes Health Indicators Dataset' from Kaggle.

Cell 3 (markdown)

Setup Kaggle API for Dataset Download

To download the dataset directly from Kaggle, you'll need a Kaggle API key. Follow these steps:

1. Go to Kaggle.com and log in.
2. Click on your profile picture (top right) and select 'Your Profile'.
3. Navigate to the 'Account' tab.
4. Scroll down to the 'API' section and click 'Create New API Token'. This will download a `kaggle.json` file to your computer.
5. Open the `kaggle.json` file. It contains your username and API key.
6. In Colab, go to the 'Secrets' tab (🔑 icon on the left panel).
7. Add a new secret. For the name, use `KAGGLE\_USERNAME` and paste your Kaggle username as the value. Do the same for `KAGGLE\_KEY`, pasting your API key as the value. Make sure 'Notebook access' is enabled for both.

Cell 4 (markdown)

Target Variable Distribution

Cell 5 (markdown)

Let's examine the distribution of our target variable, `Diabetes\_binary`, to understand if we have a balanced dataset.

Cell 6 (code)

```
# Check the distribution of the target variable
display(df["Diabetes_binary"].value_counts())
display(df["Diabetes_binary"].value_counts(normalize=True))
```

Cell 7 (markdown)

Feature Distributions and Correlations

Cell 8 (markdown)

To get a better sense of our features, let's visualize the distribution of a few key numerical features and examine the correlation matrix.

Cell 9 (code)

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set the style for the plots
sns.set_style("whitegrid")

# Plot histograms for selected numerical features
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

sns.histplot(df['BMI'], kde=True, ax=axes[0])
axes[0].set_title('Distribution of BMI')

sns.histplot(df['Age'], kde=True, ax=axes[1])
axes[1].set_title('Distribution of Age (1-13 scale)')

sns.histplot(df['MentHlth'], kde=True, ax=axes[2])
axes[2].set_title('Distribution of Mental Health Days (MentHlth)')

plt.tight_layout()
plt.show()
```

Cell 10 (code)

```
# Calculate the correlation matrix
corr_matrix = df.corr()

# Plot the correlation matrix as a heatmap
plt.figure(figsize=(15, 12))
sns.heatmap(corr_matrix, annot=False, cmap='coolwarm', fmt=".2f", linewidths=.5)
plt.title('Correlation Matrix of Features')
plt.show()
```

Cell 11 (code)

```
# Install the Kaggle API client
%pip install kaggle
```

Cell 12 (code)

```
# Set up Kaggle API credentials from Colab secrets
import os
from google.colab import userdata

os.environ['KAGGLE_USERNAME'] = userdata.get('KAGGLE_USERNAME')
os.environ['KAGGLE_KEY'] = userdata.get('KAGGLE_KEY')
```

Cell 13 (code)

```
# Download the dataset
!kaggle datasets download -d alexteboul/diabetes-health-indicators-dataset

# Unzip the dataset
```

```
import zipfile
import os

zip_file_path = 'diabetes-health-indicators-dataset.zip'
if os.path.exists(zip_file_path):
    with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
        zip_ref.extractall('./data')
    print('Dataset unzipped successfully to ./data directory.')
else:
    print(f'Zip file not found at {zip_file_path}. Please ensure the download was successful.')
```

Cell 14 (code)

```
import pandas as pd

# Load the dataset into a pandas DataFrame
df = pd.read_csv('./data/diabetes_binary_health_indicators_BRFSS2015.csv')

# Display the first 5 rows of the DataFrame
display(df.head())
```

Cell 15 (markdown)

Exploratory Data Analysis (EDA)

Cell 16 (markdown)

Let's start by examining the dataset's general information and checking for any missing values.

Cell 17 (code)

```
# Display basic information about the DataFrame
df.info()
```

Cell 18 (code)

```
# Check for missing values
display(df.isnull().sum())
```